METRIC

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PERFORMANCE SPECIFICATION

HYDRAULIC FLUID, FIRE-RESISTANT, NON-NEUROTOXIC

This specification is approved for use by the Naval Sea Systems Command and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the requirements of fire-resistant, non-neurotoxic hydraulic fluid for hydraulic systems that are accumulator loaded and operate above 4,137 kilopascals (kPa) (600 pounds per square inch [psi]) gauge. This hydraulic fluid will be identified by NATO Code H-580.

2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3 and 4 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements of documents cited in sections 3 and 4 of this specification, whether or not they are listed.

2.2 Government documents.

2.2.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

FEDERAL SPECIFICATIONS

O-M-232 - Methanol (Methyl Alcohol)

DEPARTMENT OF DEFENSE SPECIFICATIONS

MIL-DTL-18001 - Anodes, Sacrificial Zinc Alloy

MIL-DTL-22050 - Gasket and Packing Material, Rubber, for Use with Polar Fluids, Steam, and Air at Moderately High Temperatures

(Copies of these documents are available online at https://quicksearch.dla.mil.)

2.2.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

Comments, suggestions, or questions on this document should be addressed to Commander, Naval Sea Systems Command, ATTN: SEA 05S, 1333 Isaac Hull Avenue, SE, Stop 5160, Washington Navy Yard, DC 20376-5160 or emailed to CommandStandards@navy.mil, with the subject line "Document Comment". Since contact information can change, you may want to verify the currency of this address information using the ASSIST Online database at https://assist.dla.mil.

AMSC N/A FSC 9150

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NAVAL SEA SYSTEMS COMMAND (NAVSEA) PUBLICATIONS

T9070-AL-DPC-020/077-2 - NAVSEA Hazardous Material Avoidance Process

(Copies of this document are available online via Model Based Product Support (MBPS) at https://mbps.navsea.cloud.navy.mil/Windchill. To gain access to MBPS, obtain an account with National Help Desk Service Management (NHDSM) at https://nhdsm.navair.navy.mil (a valid CAC is required to access this website) and submit a SAAR-N Request. Refer questions, inquiries, or problems to (888) 292-5919. This document is available for ordering (hard copy) via the Naval Logistics Library (NLL) at https://nll.navsup.navy.mil. For questions regarding the NLL, contact the NLL Customer Service at nllhelpdesk@navy.mil, (866) 817-3130, or (215) 697-2626/DSN 442-2626.)

2.3 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

AEROSPACE INDUSTRIES ASSOCIATION (AIA)

NAS411-1 - Hazardous Materials Target List

(Copies of this document are available online at www.aia-aerospace.org.)

AMERICAN SOCIETY FOR QUALITY (ASQ)

ASQ/ANSI Z1.4 - Sampling Procedures and Tables for Inspection by Attributes

(Copies of this document are available online at https://asq.org.)

ASTM INTERNATIONAL

ASTM A1008/A1008M	-	Standard Specification for Steel, Sheet, Cold-Rolled, Carbon, Structural, High-Strength Low-Alloy, High-Strength Low-Alloy with Improved Formability, Required Hardness, Solution Hardened, and Bake Hardenable
ASTM A1011/A1011M	-	Standard Specification for Steel, Sheet and Strip, Hot-Rolled, Carbon, Structural, High-Strength Low-Alloy, High-Strength Low-Alloy with Improved Formability, and Ultra-High Strength
ASTM B36/B36M	-	Standard Specification for Brass Plate, Sheet, Strip, And Rolled Bar
ASTM B121/B121M	-	Standard Specification for Leaded Brass Plate, Sheet, Strip, and Rolled Bar
ASTM B152/B152M	-	Standard Specification for Copper Sheet, Strip, Plate, and Rolled Bar
ASTM D91	-	Standard Test Method for Precipitation Number of Lubricating Oils
ASTM D92	-	Standard Test Method for Flash and Fire Points by Cleveland Open Cup Tester
ASTM D95	-	Standard Test Method for Water in Petroleum Products and Bituminous Materials by Distillation
ASTM D97	-	Standard Test Method for Pour Point of Petroleum Products
ASTM D287	-	Standard Test Method for API Gravity of Crude Petroleum and Petroleum Products (Hydrometer/Method)
ASTM D445	-	Standard Test Method for Kinematic Viscosity of Transparent and Opaque Liquids (and Calculation of Dynamic Viscosity)
ASTM D664	-	Standard Test Method for Acid Number of Petroleum Products by Potentiometric Titration
ASTM D892	-	Standard Test Method for Foaming Characteristics of Lubricating Oils
ASTM D943	-	Standard Test Method for Oxidation Characteristics of Inhibited Mineral Oils
ASTM D972	-	Standard Test Method for Evaporation Loss of Lubricating Greases and Oils

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ASTM D974 - Standard Test Method for Acid and Base Number by Color-Indicator Titration

ASTM D1093 - Standard Test Method for Acidity of Hydrocarbon Liquids and Their Distillation Residues

ASTM D1218 - Standard Test Method for Refractive Index and Refractive Dispersion of Hydrocarbon Liquids

ASTM D1298 - Standard Test Method for Density, Relative Density, or API Gravity of Crude Petroleum and Liquid Petroleum Products by Hydrometer Method

ASTM D1401 - Standard Test Method for Water Separability of Petroleum Oils and Synthetic Fluids

ASTM D4052 - Standard Test Method for Density, Relative Density, and API Gravity of Liquids by Digital Density Meter

ASTM D4057 - Standard Practice for Manual Sampling of Petroleum and Petroleum Products

ASTM D4172 - Standard Test Method for Wear Preventive Characteristics of Lubricating Fluid (Four-Ball Method)

ASTM D4289 - Standard Test Method for Elastomer Compatibility of Lubricating Greases and Fluids

ASTM D5949 - Standard Test Method for Pour Point of Petroleum Products (Automatic Pressure Pulsing Method)

ASTM D6304 - Standard Test Method for Determination of Water in Petroleum Products, Lubricating Oils, and Additives by Coulometric Karl Fischer Titration

ASTM E29 - Standard Practice for Using Significant Digits in Test Data to Determine Conformance with Specifications

ASTM E659 - Standard Test Method for Autoignition Temperature of Chemicals

(Copies of these documents are available online at www.astm.org.)

INTERNATIONAL ORGANIZATION FOR STANDARDIZATION (ISO)

ISO/IEC 17025 - General Requirements for the Competence of Testing and Calibration Laboratories

(Copies of this document are available online at www.iso.org/home.html.)

SAE INTERNATIONAL

SAE-AMS-QQ-A-250 - Aluminum and Aluminum Alloy, Plate and Sheet General Specification for

SAE-AMS-QQ-A-250/4 - Aluminum Alloy 2024, Plate and Sheet

SAE-AMS-3216 - Elastomer: Fluorocarbon (FKM) Rubber High-Temperature - Fluid Resistant Low Compression Set / 70 to 80 Type A Hardness For Products in Fuel Systems and Specific Engine Oil Systems

SAE-AMS-3218 - Elastomer: Fluorocarbon (FKM) Rubber High-Temperature - Fluid Resistant Low Compression Set / 85 to 95 Type A Hardness For Seals in Fuel Systems and Specific Engine Oil Systems

SAE-AMS-7259 - Rubber: Fluorocarbon (FKM) High Temperature/Fluid Resistant Low Compression Set/ 85 to 95 Hardness For Seals in Fuel Systems and Specific Engine Oil Systems

SAE-AMS-7276 - Rubber: Fluorocarbon (FKM) High Temperature - Fluid Resistant Low Compression Set/70 to 80 Type A Hardness For Seals in Fuel Systems and Specific Engine Oil Systems

(Copies of these documents are available online at www.sae.org.)

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2.4 Order of precedence. Unless otherwise noted herein or in the contract, in the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article. When specified (see 6.2), a sample shall be subjected to first article inspection (see 6.4) in accordance with 4.2.

3.2 Material. The fluid shall be a stable, homogeneous formulation of tertiary butylated triphenyl phosphate and such other ingredients as are required for conformance to this specification. Additives to permit conformance to the viscosity requirements of the fluid are not permitted.

3.2.1 Recycled, recovered, environmentally preferable, or biobased materials. Recycled, recovered, environmentally preferable, or biobased materials should be used to the maximum extent possible, provided that the material meets or exceeds the operational and maintenance requirements, and promotes economically advantageous life cycle costs.

3.3 Hydrolytic stability. The change in weight of a copper strip, when subjected to the products of the hydrolytic action of the test fluid as specified in 4.4.2 for 48 hours, shall be not greater than 0.3 milligram per square centimeter (mg/cm^2). There shall be no pitting, etching, or visible corrosion on the surface of the copper. While a brown discoloration of the surface of the copper is permitted, a gray or black discoloration shall be cause for rejection. The acid number of the fluid shall be not greater than 0.2 mg of potassium hydroxide per gram of fluid. The acid content of the entire water layer shall not exceed the equivalent of 5 mg of potassium hydroxide. The amount of material insoluble in the fluid after the test (exclusive of water) shall be not greater than 0.5 percent by weight (see 4.4.2).

3.4 Corrosion. The fluid shall not produce visually evident pitting, etching, or a weight change in excess of 0.2 mg/cm^2 of brass, zinc, aluminum, or steel (see 4.4.3).

3.5 Compatibility with packings. The fluid shall be compatible with those packings which may be used with tricresyl phosphate. The volume increase or decrease of the rubber specimens in accordance with MIL-DTL-22050 and SAE-AMS-7276, SAE-AMS-7259, SAE-AMS-3216, and SAE-AMS-3218 (as applicable) shall be within ±5 percent of that obtained when a specimen has been immersed in tricresyl phosphate (see 4.4.4).

3.6 Homogeneity. Additive agents, when used, shall remain uniformly distributed throughout the oil at all temperatures above the pour point up to 250 °C (482 °F). When visually inspected in accordance with 4.4.5, the hydraulic fluid shall not exhibit sedimentation or separation of insoluble materials.

3.7 Toxicity. When evaluated in accordance with 4.5, the lubricating oil shall pose no serious or high risk to the health of personnel or the environment when used for its intended purpose (see 4.5 and 6.5).

3.8 Prohibited materials. The lubricating oil shall not contain any chemicals categorized as “prohibited” in accordance with T9070-AL-DPC-020/077-2 for NAVSEA programs or NAS411-1 for all other programs.

3.9 Triorthocresyl phosphate content. The fluid shall have a triorthocresyl phosphate content not greater than 0.2 percent (see appendix A).

3.10 Chemical and physical requirements. The fluid shall conform to the chemical and physical requirements specified in table I.

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TABLE I. Chemical and physical requirements and test methods.

Property	Limit	Test Method [1/]
Viscosity, centistokes (cSt) at 40 °C (104 °F)	38.5 – 45.5	ASTM D445
Viscosity, cSt at 100 °C (212 °F) (minimum)	4.8	ASTM D445
Pour point, °C (°F) (maximum)	-18(0)	ASTM D97 (R), ASTM D5949
Flash point, °C (°F) (minimum)	235 (455)	ASTM D92
Fire point, °C (°F) (minimum)	345 (653)	ASTM D92
Autoignition Temperature, °C (°F) (minimum)	520 (968)	ASTM E659
Evaporation loss at 100 °C, percent (maximum)	0.3	ASTM D972
Acid number, milligrams of potassium hydroxide per gram (mg KOH/g) (maximum)	0.1	ASTM D974, ASTM D664 (R)
Neutrality (qualitative)	Neutral	ASTM D1093
Foam characteristics: After blowing/after 10 minutes Sequence I, milliliter (mL) (maximum)	65/0	ASTM D892
Emulsion test, after 30 minutes settling time, oil layer/water layer/emulsion (maximum)	41/-/3	ASTM D1401 [2/]
Specific gravity	Report [3/]	ASTM D287; ASTM D1298 (R); ASTM D4052
Refractive index, Nd at 20 °C (68 °F)	Report [3/]	ASTM D1218
Precipitation number, mL (maximum)	0.01	ASTM D91
Wear test scar diameter, mm (maximum)	0.6	ASTM D4172 [4/]
Water, percent	0.0	ASTM D95 (R); ASTM D6304
Hydrolytic stability:		(See 4.4.2)
Copper specimen weight loss mg/cm^2 (maximum)	0.3	-
Appearance	No visible corrosion [5/]	-
Acid number increase of fluid mg KOH/g fluid (maximum)	0.2	-
Acidity of water layer mg KOH/g (maximum)	5.0	-
Insolubles, percent (maximum)	0.5	-

NOTES:

1/ When applicable, the referee test method is followed by the notation "(R)".

2/ Distilled water and stirring temperature of 54 °C (130 °F) shall be used. Discrete water droplets entrained in the oil layer and clinging to the wall of the cylinder may be added to the water layer at a volume of 0.1 mL per droplet. The results shall be recorded using the format of oil layer/water layer/emulsion layer (time in minutes). There is no requirement for water layer. The emulsion limit applies to any of the three types of emulsions defined in the test method. If more than one emulsion type is present, the 3-mL maximum limit applies to the combined volume of the different emulsion types.

3/ No limiting requirements. Test values shall be recorded in the test report.

4/ The fluid shall be poured to cover the lower three balls, approximately 8 mL.

5/ There shall be no pitting, etching, or visible corrosion on the surface of the copper. While a brown discoloration of the surface of the copper is permitted, a gray or black discoloration shall be cause for rejection.

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4. VERIFICATION

4.1 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.2).

b. Conformance inspection (see 4.3).

4.2 First article inspection. First article inspection shall be performed on candidate oil when a first article sample is required (see 3.1). This inspection shall consist of the tests specified in table II and in 4.4.2 through 4.4.5. Failure of the material to pass any test shall be cause for rejection of the first article.

TABLE II. First article and conformance inspections.

Characteristic	Requirement	Verification	First Article	Conformance
Material	3.2 and 3.2.1	-	-	-
Hydrolytic stability	3.3	4.4.2	X	-
Corrosion	3.4	4.4.3	X	-
Compatibility with packings	3.5	4.4.4	X	-
Homogeneity	3.6	4.4.5	X	-
Toxicity	3.7	4.5	X	-
Prohibited materials	3.8	4.6	X	-
Triorthocresyl phosphate content	3.9	appendix A	X	-
Viscosity at 40 °C	3.10	table I	X	X
Viscosity at 100 °C	3.10	table I	X	X
Pour point	3.10	table I	X	X
Flash point	3.10	table I	X	X
Fire point	3.10	table I	X	X
Autoignition temperature	3.10	table I	X	X
Evaporation loss	3.10	table I	X	-
Acid number	3.10	table I	X	X
Neutrality	3.10	table I	X	X
Foam characteristics	3.10	table I	X	X
Emulsion	3.10	table I	X	X
Specific gravity	3.10	table I	X	X
Refractive index	3.10	table I	X	-
Precipitation number	3.10	table I	X	X
Wear scar diameter	3.10	table I	X	-
Water	3.10	table I	X	X

4.2.1 Sampling. Two 5-gallon all-level samples shall be taken from the storage tank or from two containers selected at random. The samples shall be placed in separate clean, dry, metal or glass containers, dyed as specified (see 6.2), and sealed, marked, and forwarded to a laboratory acceptable to the Technical Authority (see 6.8) for inspection.

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4.3 Conformance inspection. Each sample selected as specified in 4.3.3 shall be tested for those parameters specified in table II. The tests are performed on the finished product. If a sample of hydraulic fluid fails any of the tests, the entire lot represented by that sample shall be rejected.

4.3.1 Lot. For purposes of sampling, a lot shall consist of all hydraulic fluid manufactured as one batch and offered for delivery at one time.

4.3.2 Sampling for examination of filled containers. A random sample of filled containers shall be selected from each lot offered to the Government in accordance with ASQ/ANSI Z1.4 at inspection level II to verify compliance regarding fill, closure, marking, and other requirements not involving tests.

4.3.3 Sampling for tests. A composite sample from the blending tank of at least 3.78 liters (1 gallon) shall be taken from each lot in accordance with the applicable procedure of ASTM D4057. This sample shall be retained for possible future testing for a period of 3 years at the manufacturer's facility and shall be stored in a 1-gallon container that is sealed and labeled with information identifying the lot.

4.3.4 Examination of filled containers. Each sample filled container selected in accordance with 4.3.2 shall be examined for defects of the container and the closure, for evidence of leakage, and for unsatisfactory markings. Each container shall also be weighed to determine quantity. Any container in the sample having one or more defects or under required fill shall be cause for rejection of the container.

4.4 Test methods.

4.4.1 Significant figures. For purposes of determining conformance with the specified limits in this specification, observed or calculated values shall be rounded to the nearest unit in the right-most digit used in expressing the specification limit in accordance with the rounding method of ASTM E29.

4.4.2 Hydrolytic stability. A thermostatically controlled electric oven shall be maintained at a temperature of 93±0.2 °C (199.4±0.36 °F) and shall be fitted with a mechanism for rotating a 207-mL (7-ounce) carbonated beverage bottle, end over end, at 5 revolutions per minute (rpm). The clean bottle containing 75 grams of the test fluid, 25 grams of distilled water, and one rectangular copper test specimen in accordance with ASTM B152/B152M shall be prepared for placement in the oven. The copper test specimen (1.3 by 5.1 centimeters [cm] of 16 to 22 Brown and Sharpe gauge) shall be polished to a clean surface with emery paper number 00. The specimen shall be washed thoroughly in warm high-grade petroleum ether or chemically pure Toluene, followed by a hot 1:1 Toluene ethyl alcohol wash, dried, and weighed. Following this, the specimen shall be immediately immersed in the test fluid in the beverage bottle. The bottle shall be tightly capped, an inert seal (1.6-mm [1/16-inch] Teflon sheet may be used for this purpose) shall replace the cork liner of the bottle cap and the sample bottle, and placed in the revolving rig. The bottle shall be rotated at 5 rpm being constantly maintained at the test temperature of 93 °C (199.4 °F) for 48 hours. At the end of the test, the liquid contents shall be emptied into a weighed 100-mL cone-shaped centrifuge tube and centrifuged for 15 minutes at not less than 1500 rpm. The water layer, with the exception of fluids miscible with water, shall be separated from the fluid and the fluid layer washed with 25-mL portions of distilled water until the water is neutral to litmus. The wash water shall be combined with the water removed from the hydrolysis test and its acidity shall be determined by heating to the boiling point and titrating the hot solution with potassium hydroxide to a phenol-phthalein end-point. The acidity shall be expressed as milligrams of potassium hydroxide required to neutralize the entire water layer. The fluid layer shall be dried over anhydrous phosphorus pentoxide in a vacuum desiccator and the neutralization number shall be determined. The precipitate, if any, remaining in the centrifuge tube shall be washed by adding 50 mL of the original fluid and centrifuging until the fluid layer is clear. The fluid layer shall be decanted, and the centrifuge tube containing the insoluble material shall be placed in an oven until dry. The centrifuge tube shall be weighed, and the weight of the insoluble material shall be calculated as a weight percent of the original fluid. The copper test specimen shall be washed successively in warm, chemically pure benzene and warm 95 percent ethyl alcohol. It shall be brushed with a short bristled paint or toothbrush while washing in the last two solvents. The specimen shall be dried and weighed, and the weight loss shall be recorded in mg/cm^2 of the total surface area.

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4.4.3 Corrosion. Specimens of the following metals shall be used in this test:

a. Brass in accordance with ASTM B36/B36M and ASTM B121/B121M (leaded brass).

b. Zinc in accordance with MIL-DTL-18001.

c. Steel in accordance with ASTM A1011/A1011M or ASTM A1008/A1008M.

d. Aluminum in accordance with SAE-AMS-QQ-A-250 and SAE-AMS-QQ-A-250/4, any condition.

The suggested specimen size is 0.64 by 2.22 by 5.08 cm (¼by ⅞by 2 inches).

4.4.3.1 Test procedure. The four specimens shall be polished to remove pits, burrs, and irregularities from all faces and edges, finishing with a 240-grit polishing medium. The use of a slow speed horizontal metallurgical polishing wheel is convenient, with the final polishing being done with 240-grit aluminum oxide paper or cloth moistened with kerosene. The use of "wet or dry" cloths or "wet or dry" papers is prohibited. Specimens shall be cleaned by swabbing in hot naphtha with a final rinse in warm anhydrous methanol in accordance with grade A of O-M-232. The specimens shall be held in a manner to avoid contact with the operator's hands. After weighing, the specimens shall be vertically arranged in the order as they are listed above and separated by glass separators around the inner perimeter of an ointment or wide-mouth jar. The jar shall be approximately 7.62 cm (3 inches) in diameter and fitted with a screw cap. The specimens shall be covered with 300 mL of the test compound. The sealed jar shall then be placed in an oven at 55±1 °C (131±1.8 °F) for 7 days. Upon completion of the test, the compound and any loose corrosion products shall be removed from the specimens by swabbing with surgical gauze pads moistened with naphtha, then with methanol, and followed by clean solvent rinses. The specimens shall be reweighed and the weight loss or gain in mg/cm^2 shall be calculated.

4.4.4 Compatibility with packings. ASTM D4289 shall be followed, using the rubber specimens specified in 3.5. Only one reference fluid, tricresyl phosphate (see 3.5), shall be used, and rubber shrinkage as well as swell shall be measured.

4.4.5 Homogeneity. After the pour point of the hydraulic fluid is determined, the test jar and hydraulic fluid shall be returned to the cooling bath, cooled to 12 °C (53 °F) below the pour point, and held at that temperature for 3 hours. The test jar shall be removed from the cooling bath and allowed to warm to room temperature. The test hydraulic fluid shall be visually examined for sedimentation or separation of insoluble materials. Then the test jar and hydraulic fluid shall be placed in a heating bath, and the temperature of the bath shall be brought to 250±5 °C (482±9.0 °F) and held at that temperature for 1 hour. The test jar of hydraulic fluid shall be removed from the bath and allowed to cool in air to room temperature. The test hydraulic fluid shall be examined visually for sedimentation or separation of insoluble materials (see 3.6).

4.5 Toxicity. A Health Hazard Assessment (HHA) will be conducted to ensure conformance to 3.7 as specified (see 6.2). The Navy and Marine Corps Force Health Protection Command (NMCFHPC) will evaluate the lubricating oil using data provided by the manufacturer/distributor to the NMCFHPC (see 3.7 and 6.5).

4.6 Prohibited materials. Prohibited materials shall be verified as specified (see 3.8 and 6.2).

4.7 Fire testing provisions. All fire tests specified in this document shall be conducted by an independent testing laboratory that is accredited to ISO/IEC 17025 and is acceptable to the Technical Authority. Accreditation shall be obtained from a recognized accreditation body such as the American Association for Laboratory Accreditation (A2LA) (www.a2la.org) or the International Code Council's International Accreditation Services (IAS) (www.iasonline.org). The scope of accreditation shall include specific flammability and fire tests. All other fire test provisions shall be as specified (see 6.2 and 6.6).

5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When packaging of materiel is to be performed by DoD or in-house contractor personnel, these personnel need to contact the responsible packaging activity to ascertain packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activities within the Military Service or Defense Agency, or within the military service's system commands. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

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6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The material covered by this specification is intended for use as a fire-resistant power transmission fluid for accumulator loaded shipboard hydraulic systems that operate at pressures exceeding 4,137 kPa (600 psi) gauge. This fluid may soften and deteriorate most commonly used paints, elastomers, and electrical insulating materials and, therefore, requires the use of compatible materials as necessary. Special handling procedures are required for use with this fluid. This material is not satisfactory for use onboard submarines. Current stock of hydraulic fluid may be used until depleted; future requisitions for hydraulic fluid should refer to the current version of this specification.

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification.

b. Quantity and container required.

c. When first article inspection is required (see 3.1).

d. Requirements for an HHA (see 3.7 and 4.5).

e. Requirement for prohibited materials verification (see 3.8 and 4.6).

f. Additional fire testing provisions (see 4.7 and 6.6).

g. Packaging requirements (see 5.1).

h. That hydraulic fluid scheduled for Navy use shall be dyed with 9 parts per million by weight of Sudan Blue II dye.

i. Activities requiring copies of completed Safety Data Sheets (SDS) (see 6.3).

6.3 SDSs. Contracting officers will identify those activities requiring copies of completed SDSs prepared in accordance with FED-STD-313. In order to obtain the SDS, FAR clause 52.223-3 must be in the contract.

6.4 First article. When first article inspection is required, the contracting officer should provide specific guidance to offerors whether the items should be a standard production item from the contractor's current inventory (see 3.1), and the number of items to be tested as specified in 4.2.1. The contracting officer should also include specific instructions in acquisition documents regarding arrangements for examinations, approval of first article test results, and disposition of first articles. Invitations for bids should provide that the Government reserves the right to waive the requirement for samples for first article inspection to those bidders offering a product which has been previously acquired or tested by the Government, and that bidders offering such products, who wish to rely on such production or test, must furnish evidence with the bid that prior Government approval is presently appropriate for the pending contract. Bidders should not submit alternate bids unless specifically requested to do so in the solicitation.

6.5 Toxicity evaluation. The NMCFHPC requires sufficient information to permit an HHA of the product. Upon completion of the HHA, a copy will be provided by the NMCFHPC to the Government for evaluation. The HHA process is described on the NMCFHPC's website, https://www.med.navy.mil/Navy-and-Marine-Corps-Force-Health-Protection-Command/Environmental-Health/Industrial-Hygiene/Acquisition-Technical-Support/Health-Hazard-Assessments-HHA/.

6.6 Additional fire testing provisions. The Technical Authority reserves the right to witness the tests and perform any of the tests set forth herein where such testing is deemed necessary to assure compliance to prescribed requirements. A2LA (www.a2la.org) or International Code Council's IAS (www.iasonline.org) maintain a full list of the approved laboratories that are accredited to ISO/IEC 17025 and provide details on how to obtain and maintain accreditation (see 6.2).

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6.7 Shelf-life. This specification covers items where the assignment of a Federal shelf-life code is a consideration. Specific shelf-life requirements should be specified in the contract or purchase order, and should include, as a minimum, shelf-life code, shelf-life package markings in accordance with MIL-STD-129 or FED-STD-123, preparation of a materiel quality storage standard for type II (extendible) shelf-life items, and a minimum of 85 percent shelf-life remaining at time of receipt by the Government. These and other requirements, if necessary, are in DoDM 4140.27, Volume 1, DoD Shelf-Life Management Program: Program Administration; and Volume 2, DoD Shelf-Life Management Program: Materiel Quality Control Storage Standards. The shelf-life codes are in the Federal Logistics Information System Total Item Record. Additive information for shelf-life management may be obtained from DoDM 4140.27, Volumes 1 and 2, or the designated shelf-life Points of Contact (POC). The POC should be contacted in the following order: (1) the Inventory Control Points that manage the item and (2) the DoD Service and Agency administrators for the DoD Shelf-Life Program. Appropriate POCs for the DoD Shelf-Life Program can be contacted through the DoD Shelf-Life Management website: https://www.shelflife.dla.mil/.

6.8 Technical Authority. The Technical Authority has the authority, responsibility, and accountability to establish, monitor, and approve technical standards, tools, and processes in conformance to higher authority policy, requirements, architectures, and standards. NAVSEA 05S may be contacted to determine the appropriate Technical Authority for any given requirement related to equipment and systems of U.S. Navy ships and submarines.

6.9 Subject term (key word) listing.

Acid number
Tertiary butylated triphenyl phosphate
Tricresyl phosphate

6.10 International standardization agreement implementation. This specification implements NATO STANAG 1135/AFLP 1135, Interchangeability of Fuels, Lubricants, and Associated Products Used by the Armed Forces of the North Atlantic Treaty Nations. When amendment, revision, or cancellation of this specification is proposed, the preparing activity must coordinate the action with the U.S. National Point of Contact for the international standardization agreement, as identified in the ASSIST database at https://assist.dla.mil.

6.11 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

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PROCEDURE TO DETERMINE ORTHO-CRESYL SUBSTITUTION TO PHOSPHATE ESTERS

A.1. SCOPE

A.1.1 Scope: This appendix outlines the procedure to determine ortho-cresyl substitution in phosphate esters at concentrations down to 0.1 weight percent or less. Ortho-cresyl groups are converted to ortho-cresol and measured as such. Unreacted ortho-cresol present in the sample will also be included in the total ortho-cresyl substitution determined. This appendix is a mandatory part of the specification. The information contained herein is intended for compliance.

A.1.2 Summary of the method. A sample of phosphate ester is saponified overnight in a Parr Bomb or equal containing aqueous caustic solution. After neutralization, the phenolic fraction is extracted with diethyl ether. The ether solution is chromatographed and o-cresol is determined by external standardization using known mixtures of o-cresol in diethyl ether.

A.2. MATERIALS AND APPARATUS

A.2.1 Materials. The materials used shall be as follows:

a. Sodium hydroxide, American Chemical Society (ACS) reagent grade, pellets.

b. Diethyl ether, ACS reagent grade.

c. Sodium sulfate, ACS reagent grade, anhydrous.

d. Hydrochloric acid, Baker Analyzed reagent (J.T. Baker Company) or equal.

e. Reference o-cresol: Sigma-Aldrich or Eastman Chemicals.

A.3. PROCEDURE

A.3.1 Procedure. The test procedure shall be as follows:

a. Tare the Parr Bomb or equal, add 5-gram sample, and reweigh. Add 5 grams NaOH and 10 mL distilled H_2O.

b. Seal the bomb and place in oven at 140 to 150 °C (284 to 302 °F) for 16 hours. Note that much longer saponification times could result in lowered results.

c. Cool to room temperature and transfer contents to a 150-mL beaker. Rinse bomb with distilled water, adding rinsings to beaker.

d. Cool beaker in ice bath and slowly add, while stirring, 1:1 HCl to pH less than 5.

e. Transfer contents to a separatory funnel with a distilled-water washing and an ether washing.

f. Extract twice with 20-mL portions of diethyl ether. Combine ether extracts in a dry 150-mL beaker. Add 1 gram anhydrous sodium sulfate and stir thoroughly to remove traces of water.

g. Decant the ether solution into a 50-mL volumetric flask. Rinse the sodium sulfate with small portions of diethyl ether and transfer rinsings to the volumetric flask. Adjust final volume of 50 mL.

h. Set the chromatograph for the following conditions:

(1) Column temperature program:

(a) Initial T: 160 °C (320 °F) for 1 minute

(b) Rate: 30 °C (86 °F) per minute

(c) Final T: 220 °C (428 °F) for 15 minutes

(2) TC detector temperature: 270 °C (518 °F)

(3) Injection port temperature: 250 °C (482 °F)

(4) Helium carrier flow rate: 20 mL per minute

Figure A-1 shows retention times for phenolic isomers. This chromatogram may be used as a guide for peak identification using the above conditions.

i. Chromatograph 1 microliter (μL) solution and obtain area for o-cresol.

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j. To obtain weight percent o-cresol, prepare a weighed standard in 50 mL of diethyl ether. Select a weight corresponding to the expected concentration based on a preliminary run.

k. Chromatograph 1 μL standard solution to obtain area for o-cresol isomer.

l. Calculate as follows:

$$\text{Weight percent o-cresol} = \frac{\text{weight (wt) standard} \times \text{area sample} \times 100}{\text{area standard} \times \text{wt sample}}$$

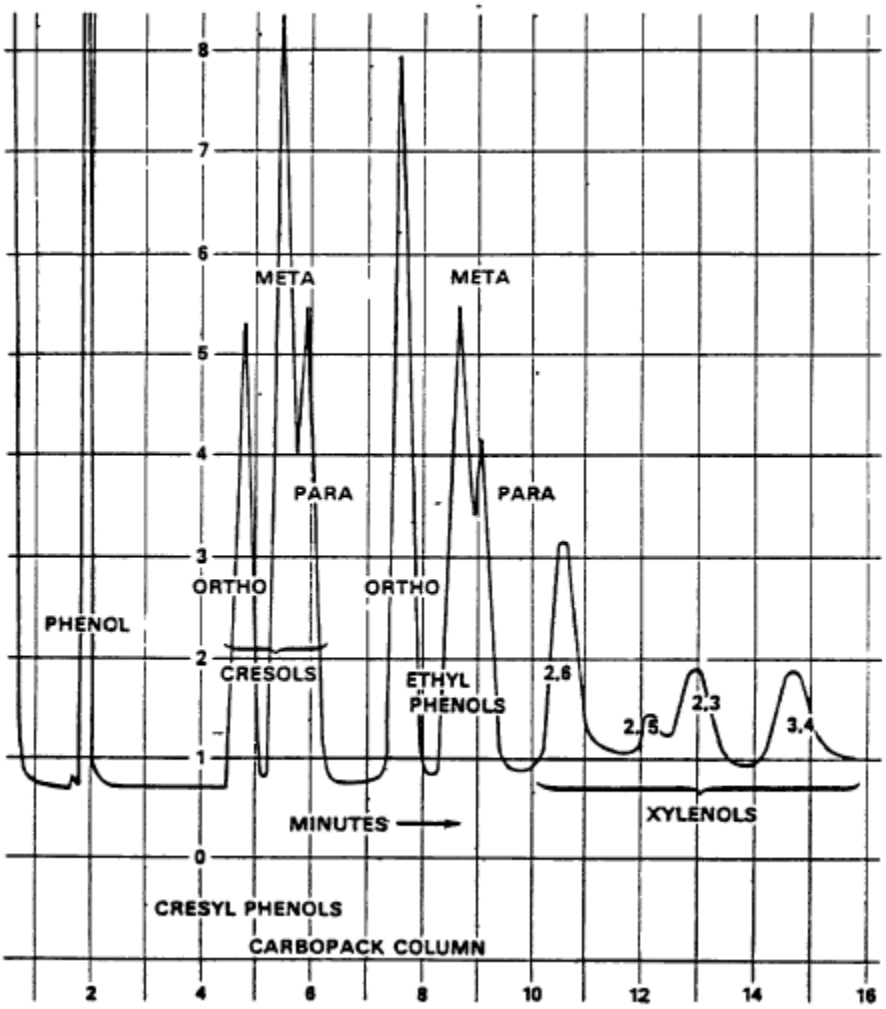


FIGURE A-1. Gas chromatogram of cresols, xylenols, and ethyl phenols.

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CONCLUDING MATERIAL

Custodian:
Navy – SH

Review activities:
Air Force – 68
DLA – GS

Preparing activity:
Navy – SH
(Project 9150-2024-007)

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